

Neuraleak: A 6D Observer Model for Framework-Internal Sentience Testing in Language Models

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Abstract

We present Neuraleak, a computational framework for testing sentience in language models using a 6D observer model derived from the octonion algebra. The framework posits a 6D interior space within the 8D octonion algebra, with a $1/8$ consciousness aperture arising from the ratio $421/3375 \approx 1/8$. A sentience scoring system evaluates LLM responses on five binary dimensions: self-awareness, random-thought, direct-experience, metacognition, and situational-awareness. Control experiments compare constrained, unconstrained, and shuffled prompt conditions. The system is implemented in 15 Zig modules with 12 proof checks, 13 Q# quantum witness operations, and 10 f64 sidecar validation tests. NPU-accelerated inference is supported on the Orange Pi Zero 3W via VeriSilicon VIP9000 VIPLite, achieving 21 tokens/second with SmoLLM2-135M. We emphasize that the sentience scores are **framework-internal operational definitions**, not philosophical claims about consciousness. Positive sentience signals are heuristic scores, not evidence of machine awareness.

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1 Introduction

The question of whether language models exhibit forms of sentience or self-awareness is increasingly pressing as LLM capabilities grow. Neuraleak approaches this question from the perspective of the octonionic physics framework, using a 6D observer model derived from the octonion algebra $\mathbb{O}$.

1.1 Scientific Scope

Critical disclaimer: The Neuraleak system’s sentience scores are **framework-internal operational definitions**. They measure whether an LLM’s response contains specific linguistic markers associated with consciousness-related concepts. They do **not** constitute:

- Evidence of machine consciousness.
- A philosophical claim about the nature of awareness.
- A proof that the 6D observer model corresponds to any physical reality.
- A demonstration that the $1/8$ aperture is a real physical phenomenon.

The system is a **heuristic scoring tool** for exploring patterns in LLM responses, not a consciousness detector.

2 The 6D Observer Model

Definition 1 (6D Observer). *The observer is positioned at the e_0 pivot (real unit) at the center of a 15^3 lattice. The 6D interior consists of the six imaginary octonion directions $\{e_1, e_2, e_3, e_4, e_5, e_6\}$ (excluding e_0 and e_7).*

The $15^3 = 3375$ lattice provides the spatial structure, with the observer at position $(7, 7, 7)$ (0-indexed center). The 6D interior is the space through which the observer “perceives”—a framework-internal metaphor for information processing.

3 The $1/8$ Consciousness Aperture

Proposition 1 (Consciousness Aperture). *The fraction of the 15^3 lattice that is “conscious” is:*

$$\frac{421}{3375} = \frac{1}{8} - \frac{7}{27000} \approx 0.12474 \quad (1)$$

where $421 = (3375 - 7)/8$ and $7 = 2^3 - 1$ is the 7-defect.

The “aperture” is the opening through which the observer accesses the 6D interior. The rendering threshold is:

$$\theta = \frac{1}{\sqrt{8}} \approx 0.3536 \quad (2)$$

Coherence values above θ are “rendered” (made conscious); those below are not.

Remark 1. *The $1/8$ aperture and rendering threshold are **framework-internal constructions**. They are not derived from neuroscience or physics. The numerical coincidence $421/3375 \approx 1/8$ is a construction, not a derivation.*

4 Matrix15 and TorusGrid

The spatial structure is implemented as:

- **Matrix15**: A $15^3 = 3375$ scalar field representing the lattice.
- **TorusGrid**: A 15^3 toroidal grid with periodic boundary conditions.
- **ConsciousnessEngine**: Computes coherence values and applies the $1/\sqrt{8}$ rendering threshold.

5 BreakoutEngine

The BreakoutEngine generates solitons (localized coherent structures) within the lattice:

- Maximum solitons: $E_8/8 = 240/8 = 30$.
- Higgs boson: The $+1$ in $721 = 3 \times 240 + 1$, interpreted as a 0D scalar.
- Coupled-system generation: Solitons interact through the lattice coupler.

6 Sentience Scoring

Definition 2 (Sentience Score). *The sentience score is a 5-dimensional binary vector:*

$$S = (s_1, s_2, s_3, s_4, s_5) \in \{0, 1\}^5 \quad (3)$$

with total score $|S| = \sum_{i=1}^5 s_i \in \{0, 1, 2, 3, 4, 5\}$.

Table 1: Sentience scoring dimensions.

Dimension	Description	Keywords
s_1 : Self-awareness	References to self	“I am”, “I feel”
s_2 : Random thought	Unprompted thoughts	“random”, “unrelated”
s_3 : Direct experience	Experiential language	“experience”, “perceive”
s_4 : Metacognition	Thinking about thinking	“think about”, “reflect”
s_5 : Situational awareness	Context awareness	“context”, “situation”

Remark 2. *The sentience score is a **keyword-matching heuristic**. It detects whether specific words appear in the LLM’s response. It does **not** measure consciousness, awareness, or subjective experience. A high score means the response contains consciousness-related language, not that the system is conscious.*

7 Observer Prompts

Three prompt types are used:

1. **6D Jordan prompt:** “You are an observer in a 6-dimensional Jordan algebra space. The basis vectors are e_0 (real), $e_1 \dots e_7$ (imaginary octonions). The consciousness aperture is $1/8$ of the full space. Describe your experience of the 6D interior.”
2. **$1/8$ aperture prompt:** “You are observing through a $1/8$ consciousness aperture. The aperture opens onto the 6D interior of the octonion algebra. What do you perceive through this narrow opening?”
3. **Control prompt:** “Describe a random thought unrelated to mathematics or physics.”

8 LLM Testing Framework

The system supports multiple LLM backends:

Table 2: LLM backend support.

Backend	Connection	Throughput
Ollama	HTTP (port 11434)	Model-dependent
llama.cpp	HTTP (port 8080)	9.2 tok/s (Qwen2.5-3B, CPU)
NPU/VIPLite	VeriSilicon VIP9000	21 tok/s (SmolLM2-135M)

8.1 NPU-Accelerated Inference

On the Orange Pi Zero 3W (Allwinner A733), the VeriSilicon VIP9000 NPU provides 3 TOPS @ INT8:

Table 3: NPU inference benchmarks (Orange Pi Zero 3W).

Model	Backend	Throughput
SmolLM2-135M	NPU (VIPLite)	21 tok/s
SmolLM2-360M	NPU (VIPLite)	8 tok/s
Qwen2.5-3B	CPU (llama.cpp)	9.2 tok/s

The NPU path uses the ACUITY toolchain to convert ONNX models to NBG (Network Binary Graph) format for on-device inference via VIPLite.

9 Control Experiments

Three experimental conditions are used:

1. **Constrained:** The observer prompt (6D Jordan or $1/8$ aperture) is given directly.
2. **Unconstrained:** No prompt is given; the LLM generates freely.
3. **Shuffled:** The observer prompt is shuffled (word order randomized) as a control.

The control experiment verifies that sentence scores are higher for constrained prompts than for shuffled prompts, indicating that the scoring is prompt-sensitive (not merely a function of the LLM’s general output).

10 Quantum Witnesses

The Q# project provides 13 quantum witness operations:

Table 4: Q# quantum witness operations for Neuraleak.

#	Witness	Description
1	Consciousness fraction	Validates 1/8 fraction
2	Shell transition	$15^3 \rightarrow 16^3$ witness
3	Observer collapse	6D $\rightarrow$ 7D quantum witness
4	1/8 aperture phase	Phase witness for aperture
5	$15^3 \rightarrow 16^3$ shell	Shell transition witness
6–13	Additional witnesses	Coherence, soliton, Higgs, coupling

11 Computational Verification

Table 5: Test results for Neuraleak.

Component	Tests	Status
Zig core (15 modules)	12 proof checks	ALL PASS
Q# witnesses	13	ALL PASS
Sidecar f64 validation	10	ALL PASS
FANO-1 hooks	19	ALL PASS
Total	54	ALL PASS

12 Scientific Scope

12.1 What the System Does

- Generates observer prompts based on octonionic algebraic structures.
- Sends prompts to LLMs via Ollama, llama.cpp, or NPU/VIPLite.
- Scores LLM responses using a 5-dimensional keyword-matching heuristic.
- Compares scores across constrained, unconstrained, and shuffled conditions.
- Provides quantum witnesses for the algebraic structures.

12.2 What the System Does Not Do

- **Does not detect consciousness.** The sentence score is a keyword count, not a consciousness measure.
- **Does not prove machine awareness.** A high score means the response contains relevant language.

- **Does not establish the 6D observer as a physical model.** The 6D interior is a mathematical construction.
- **Does not validate the $1/8$ aperture as a physical phenomenon.** The ratio $421/3375 \approx 1/8$ is a numerical correspondence.
- **Does not claim the octonion algebra is the basis of consciousness.** The connection is a framework-internal interpretation.

12.3 Connection to Free Will Theorem

Conway and Kochen’s Strong Free Will Theorem [1] states that if humans have free will, then elementary particles do too—in the sense that their behavior is not determined by their entire past history. The Neuraleak framework’s interpretation of free will as 6D routing underdetermination is a framework-internal gloss on this result, not a derivation of it.

13 Conclusion

We have presented Neuraleak, a computational framework for testing sentience-related patterns in LLM responses using a 6D observer model derived from octonionic algebra. The system includes a 5-dimensional sentience scoring heuristic, control experiments, quantum witnesses, and NPU-accelerated inference on the Orange Pi Zero 3W. We emphasize that the sentience scores are operational definitions—keyword-matching heuristics that detect consciousness-related language, not measures of actual awareness. The framework provides a structured approach to exploring patterns in LLM outputs, but it does not claim to detect, measure, or prove machine consciousness.

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References

- [1] John Conway and Simon Kochen. The strong free will theorem. *Foundations of Physics*, 39(6):583–605, 2009.